NATO field-trial SERS dataset

Progress update, results and paper options

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Meeting with Li-Lin · 21 October 2026

Since we last spoke (14 September)

Already reported: the classical machine-learning baseline. Extra Trees identified the chemical on an instrument it had never seen with 68.5% balanced accuracy (chance: 33%).

New since then

1. Compact neural networks (CNNs), including versions trained to ignore instrument differences.
2. Three preprocessing pipelines — minimal, smoothing, baseline correction — tested on every model.
3. Your question from 1 September: do the substrates detect the analyte independently of the instrument?

For discussion

- Paper 1: what this dataset supports, and what it still needs.
- Paper 2: the original VAE goal, and what a proper dataset for it looks like.

What we set out to do

Original idea

Train a disentangled variational autoencoder (VAE) to split each spectrum into a chemical part and a substrate/instrument part.

Why this dataset cannot support it

- No pure-analyte reference spectra, so a "chemical part" cannot be checked.
- Chemicals, substrates and instruments are not crossed: 374 of 2,760 sample $\times$ substrate $\times$ instrument combinations were measured.
- Only 69 independently prepared samples.

Current focus

Your clarification: the trial asked whether the SERS substrates detect the analyte *independently of the instrument*.

Main question

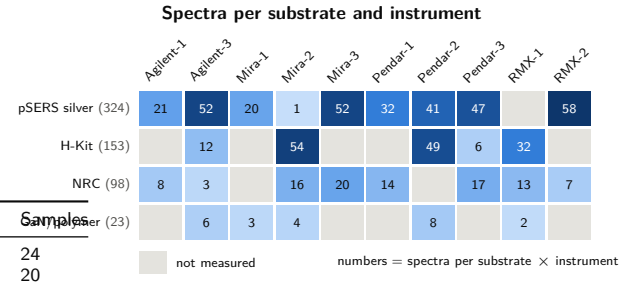
Can a model identify the chemical on an instrument — and on samples — that it never saw during training?

Follow-up questions

When does preprocessing help? Do neural networks add anything over classical methods? Which substrates hold up across instruments?

The dataset

1,172 logged recordings
721 with a named SERS substrate
626 pass numeric quality checks
615 traceable to a sample and target
598 unique spectra
69 independent physical samples

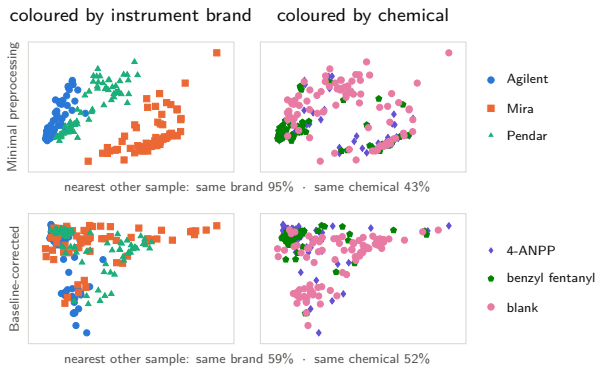


Station	Chemicals	Samples
CWA	4-nitrophenol, ethanol, ethyl paraoxon	24
Pills	4-ANPP, benzyl fentanyl, blank	20
Surfaces	4-ANPP, acetaminophen, benzyl fentanyl	25

Two reduced sets check the results: 500 spectra (quality-flagged removed) and 575 (Mira-1 removed; it failed its system check).

The 598 spectra are repeat views of 69 samples. Every test keeps all views of one sample on the same side of the train/test split.

Instrument differences dominate the raw spectra



Pills station: principal-component map of 208 spectra (one point per spectrum).

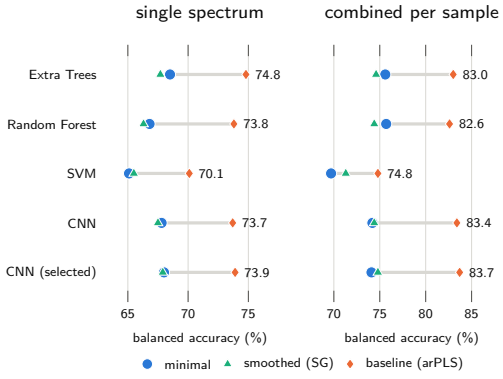
Minimal preprocessing: a spectrum's nearest neighbour from another sample has the same instrument brand 95% of the time, but the same chemical only 43%. Both would be about 1 in 3 by chance.

Baseline correction weakens the brand grouping (59% brand, 52% chemical). At Surfaces it reverses: 42% brand, 74% chemical.

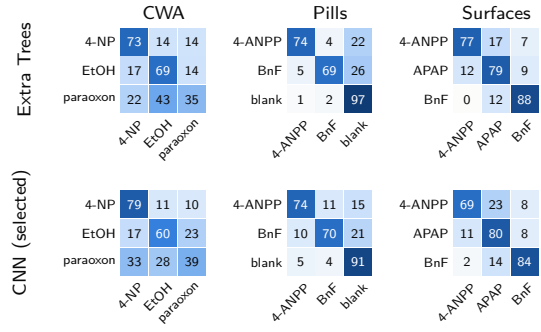
Removing the background moves spectra from grouping by instrument toward grouping by chemical. It does not remove the instrument effect completely.

Results: identifying chemicals on an unseen instrument

Five models × three preprocessing pipelines



Which chemicals are confused



rows = true chemical, columns = predicted;
% of each true chemical; baseline-corrected

Baseline correction helps every model; tree models and CNNs perform alike. With minimal preprocessing, Extra Trees calls 42–45% of Pills drug spectra "blank"; correction roughly halves this. Ethyl paraoxon is the hardest chemical (35–39% correct).

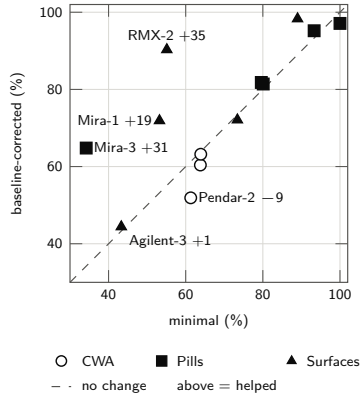
"Combined per sample" averages the model's probabilities over a sample's repeat spectra (spectra are not averaged). 4-NP = 4-nitrophenol, EtOH = ethanol, BnF = benzyl fentanyl, APAP = acetaminophen.

Results: where it works and where it does not

Per chemical, all models (% correct)

		Extra Trees	Rand. Forest	SVM	CNN	CNN (sel.)
CWA	4-nitrophenol	73	73	69	78	79
	ethanol	69	68	67	60	60
	ethyl paraoxon	35	28	37	39	39
Pills	4-ANPP	74	73	72	74	74
	benzyl fentanyl	69	69	66	70	70
	blank	97	93	97	91	91
Surfaces	4-ANPP	77	71	65	68	69
	acetaminophen	79	80	73	80	80
	benzyl fentanyl	88	88	75	83	84

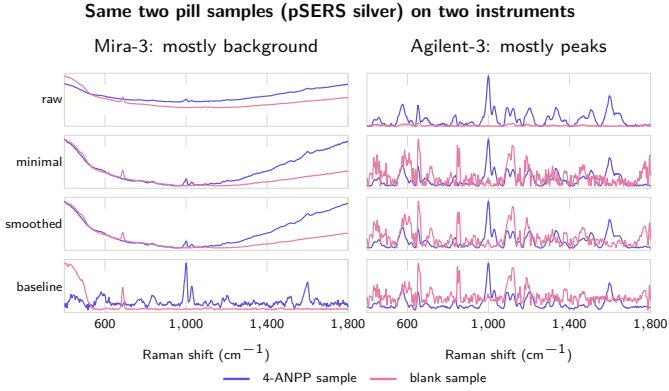
Per instrument (Extra Trees)

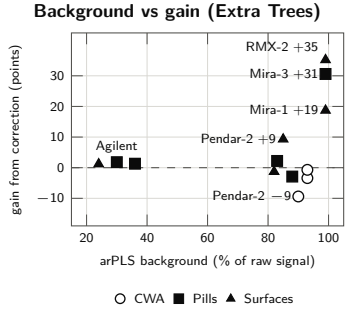


Blank is easy (91–97%) and ethyl paraoxon is near chance (28–39%); the models agree chemical by chemical. Baseline correction rescues background-heavy instruments (Mira-3 +31, RMX-2 +35 points) but lowers every CWA instrument (by 1–9 points).

Both panels: baseline-corrected, unseen instrument. Right: each point is one station–instrument combination; points above the dashed line improved with baseline correction.

Why baseline correction helped some instruments and not others





On Mira-3, drug and blank spectra look alike until the background is removed (median correlation 0.94 → 0.16): this is why drugs were called "blank" on slide 6. Correction pays off where the raw signal is almost all background (≈99%: RMX-2, Mira-1, Mira-3; +19 to +35 points). Below that, results are mixed.

Samples: lowest-ID 4-ANPP and blank pills on both instruments, first scan each; correlations use all Pills/pSERS spectra. Right: background = area the models' own arPLS step removed ÷ raw area. Exploratory (examined after the results); rank correlation with gain 0.73–0.75 for SVM and CNNs, 0.26–0.30 for tree models.

Paper 1: what this dataset supports now

Working title: Chemical identification on unseen handheld Raman instruments: a leakage-controlled benchmark on a multi-instrument SERS field trial

The gap: no published study has tested whether a SERS model still identifies the chemical on a handheld instrument it has never seen.

Contributions

- Curated field-trial dataset with its coverage matrix.
- Strict test: unseen instrument *and* unseen samples.
- Tree models and compact CNNs perform alike.
- Baseline correction: $+5$ to $+7$ points, but instrument-dependent.
- Substrate performance across instruments is condition-specific.

Closest prior work

- Fornasaro et al. and Guo et al., *Anal. Chem.* 2020: spectra differ across labs and devices — variability measured, identification not benchmarked.
- Mahmud et al., *Spectrochim. Acta A* 2026: maps 4 portable SERS instruments to a lab instrument — bacteria, one substrate.
- Lai et al., *Anal. Chem.* 2025; She et al., *Anal. Chem.* 2026: transfer models between devices — need data from the new instrument.
- Mirsafavi et al., *Analyst* 2020; Wilson et al., *Sens. Actuators B* 2021: SERS of fentanyl and 4-ANPP — one instrument.

Candidate journals

Analyst (first choice) · Analytical Chemistry (reach) · Applied Spectroscopy · Analytica Chimica Acta · J. Raman Spectrosc.; Scientific Data if the data can be released

The contribution is the test protocol and a rare dataset, not a new method. Classical $\approx$ CNN agrees with recent Raman benchmarks (Sineesh & Kamsali, *Digital Discovery* 2026). Checked 20 October 2026 against Crossref; full list in backup.

Paper 1: what it still needs

Addition	Question it answers
Calibration with 1–2 samples from the new instrument	How many samples fix a new instrument? Prior work uses target samples (Lai 2025, She 2026), so this places our results beside theirs
Other normalizations (SNV, vector, area)	Why min–max scaling?
Without Mira-1; without quality-flagged spectra	Do suspect measurements drive the results?
Why CWA got worse with correction	All three CWA instruments dropped 1–9 points
Background mechanism (slide 8)	Why correction helps some instruments and not others

Already answered: spike removal is not the cause — the smoothing pipeline also removes spikes and gained nothing.

Parked: preprocessing chosen per spectrum quality (supported in 54 of 260 tests; ≈1.6M fits); per-brand preprocessing (untestable: instruments were tied to stations); unknown-chemical rejection; extra CNN losses; RBF networks and self-organizing maps (after submission).

Plan: about one week of experiments on the existing splits and code, then freeze the results and write.

Paper 2: the original VAE goal, with a designed dataset

Working title: Separating analyte, substrate and instrument contributions in SERS spectra with structured variational autoencoders

The gap: no SERS dataset measures every chemical on every substrate and instrument, so the chemical's signal has never been separated from the rest.

What the model separates

1. Instrument response — removed.
2. Substrate background — removed.
3. The analyte's shared SERS signature — **the real signal**.
4. A substrate-specific change to the analyte (chemistry $\times$ substrate) — measured.

How it is tested

- Recover each signature; compare with pure Raman to see how much SERS alters it.
- Predict combinations deliberately left out of training.
- Compare against the Paper 1 methods.

Closest prior work

- Fornasaro et al., *Anal. Chem.* 2020: 15 labs, 6 SERS methods — but one analyte; authors caution against extrapolating to other substrates.
- Mahmud et al., *Spectrochim. Acta A* 2026: 20 analytes, 5 instruments — but one substrate.
- Abram & McCloskey, *Biomolecules* 2023: a VAE separates instrument type — in mass spectrometry.
- Lotfollahi et al., *Mol. Syst. Biol.* 2023: predicts unseen drug $\times$ cell-type combinations — a template from biology.

Candidate journals
Analytical Chemistry (ACS) · Digital Discovery (RSC) · Machine Learning: Science and Technology (IOP) · Nature Communications (stretch) · plus a Scientific Data descriptor for the dataset

Paper 1 comes first: it shows acquisition effects dominate and why separating them needs a crossed design. Literature checked 20 October 2026; repeat in Scopus before submission.

Paper 2: experiments and metrics

Experiment	Question	How we measure it	Compared with
E1 Recover each analyte's SERS signature	Does the chemical part capture what the analyte shares across substrates, and how much does each substrate alter it?	Match to the shared signature; comparison with pure Raman shows how much SERS alters it	Standard unmixing (MCR-ALS, NMF)
E2 Separate chemical from substrate	Does the chemical part hold only chemistry?	Chemical readable from the chemical part, not from the substrate part	The model ladder below
E3 Predict unmeasured combinations	Can it predict a chemical on a substrate it never saw together?	Match to the real spectrum	Simple additive model
E4 Identify on a new substrate	Can it identify chemicals on a substrate left out of training?	Balanced accuracy	Paper 1 methods

Model ladder (E2): VAE $\rightarrow$ β-VAE (unsupervised disentanglement) $\rightarrow$ conditional VAE with label-partitioned latents (supervised disentanglement) $\rightarrow$ physics-structured VAE with an additive decoder (ours). Where it starts working shows what separation needs.

Rules: always split by prepared sample; decide what counts as success before running.

Instrument translation and ablations in backup.

Paper 2: what a proper dataset needs

Today: independent samples per chemical × substrate

	pSERS	H-Kit	NRC	GaN
4-nitrophenol		8	5	4
ethanol		9	4	
ethyl paraoxon		7	3	
4-ANPP	11		7	
benzyl fentanyl	15		7	1
acetaminophen	11	1	4	
blank	8		8	

never measured together

1. Fill the matrix. Every analyte on every substrate and instrument, about 5 independent preparations per cell. Today H-Kit met only CWA chemicals and pSERS only Pills/Surfaces chemicals, so substrate and chemical cannot be separated.

2. Ground truth for each effect. Pure-analyte Raman references, substrate-only blanks, instrument intensity standards (e.g. NIST SRM 2241).

3. Enough instruments. $\geq$5–6, including $\geq$3 units of one brand, so "works on a new instrument" is tested more than a few times.

4. The same preparation on every instrument. The pairing shows the model directly what an instrument effect looks like.

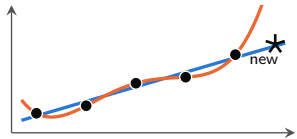
Illustrative design: (6 analytes $\times$ 3 concentrations + 1 blank) $\times$ 4 substrates $\times$ 5 preparations $=$ 380 independent samples, each read on 5 instruments $\times$ 3 scans $\approx$ 5,700 spectra.

Theory agrees: without labelled, crossed factors, disentanglement cannot be learned (Locatello et al., ICML 2019); correlated factors stay entangled (Träuble et al., ICML 2021).

Paper 2: how much data?

Like a calibration curve: a model needs more data points than adjustable parameters to give one stable answer.

(N samples $\times$ M outputs $>$ W weights)



Today 69 samples. Our CNN had thousands of times more weights than data. It worked, but never beat Random Forest.

Paper 2 ≈300–500 independent preparations, each on every substrate and instrument.

First A ≈100-preparation pilot to confirm the number.

Rule: Q.J. Zhang & K.C. Gupta, *Neural Networks for RF and Microwave Design* (Artech House, 2000). Numbers and test-precision details in backup.

For discussion

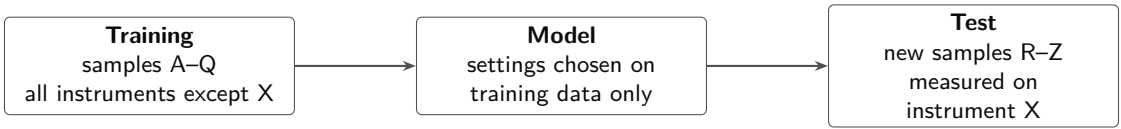
1. **Publication.** Can results from the NATO trial data be published? Who must approve, who are co-authors, and is there a citable trial report (NATO SET-292, Aberdeen, March 2024)?
2. **Route.** Paper 1 now with this dataset; Paper 2 requires new measurements.
3. **New data.** Which analytes, substrates and instruments could NRC measure? Are $\geq$5 instruments, with $\geq$3 of one brand, available? Can one SERS chip be read on several instruments? Could we start with a $\approx$100-preparation pilot?

Backup: seven questions and where each stands

	Question	Status
✓	1. Identify the chemical on an unseen instrument?	65–68% minimal; 70–75% baseline-corrected (chance 33%)
✓	2. Which single preprocessing pipeline helps most?	Baseline correction: +5 to +7 points; smoothing: no gain
~	3. Should preprocessing adapt to the instrument?	By brand: not testable here. By spectrum quality: possible at CWA only; not yet run
✓	4. Do neural networks beat classical methods?	No clear difference
○	5. How much do 1–2 calibration samples from a new instrument help?	Not yet run; feasible for 1–2 samples per chemical
○	6. Are confidence scores reliable; can unknowns be flagged?	Not yet run; CNN confidence is poorly calibrated
~	7. Do substrates work independently of the instrument?	Partly: depends on the instrument; Mira units are the weakest

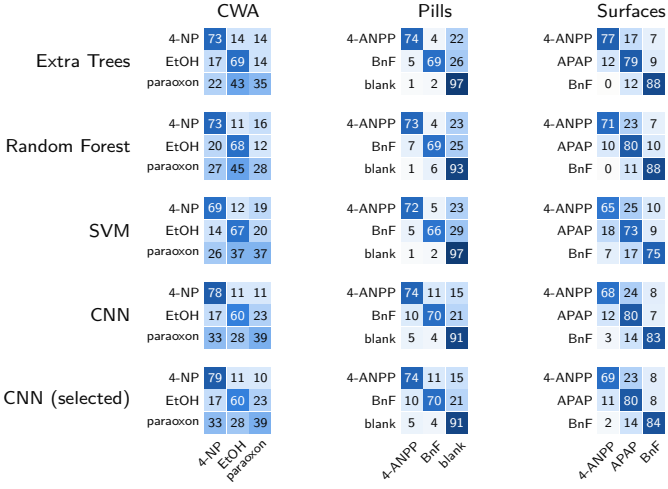
✓ answered ~ partly answered ○ not yet run. Balanced accuracy averages the hit rate over chemicals, so 33% is chance for three chemicals.

Backup: how the test was kept fair



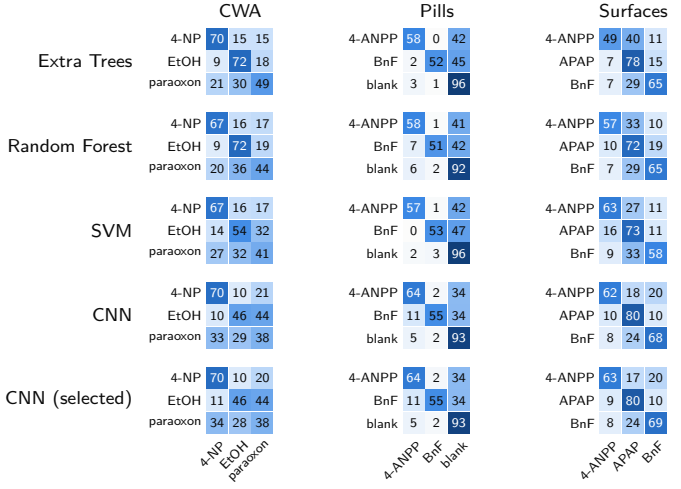
- Every spectrum of one physical sample stays on one side of the split (four folds, repeated five times).
- The held-out instrument contributes nothing to training, preprocessing statistics, model selection, calibration or stopping.
- 13 station–instrument combinations have enough samples for this test (CWA: 3, Pills: 5, Surfaces: 5).
- Scores use the same method as the 14 September report, so the 68.5% Extra Trees figure is unchanged.

Backup: all confusion matrices, baseline-corrected



Rows = true chemical, columns = predicted; % of each true chemical.

Backup: all confusion matrices, minimal preprocessing



Rows = true chemical, columns = predicted; % of each true chemical.

Backup: accuracy per instrument (%), minimal → corrected

Station	Instrument	Extra Trees	Random Forest	SVM	CNN	CNN (selected)
CWA	Mira-2	64 → 60	58 → 58	50 → 60	50 → 60	50 → 60
CWA	Pendar-2	61 → 52	60 → 48	52 → 50	49 → 59	49 → 59
CWA	RMX-1	64 → 63	63 → 63	60 → 63	54 → 60	55 → 60
Pills	Agilent-1	80 → 81	79 → 83	81 → 81	80 → 79	80 → 79
Pills	Agilent-3	93 → 95	88 → 93	96 → 92	93 → 87	93 → 87
Pills	Mira-3	34 → 65	34 → 60	33 → 64	38 → 69	38 → 69
Pills	Pendar-1	80 → 82	76 → 83	78 → 80	83 → 80	83 → 80
Pills	Pendar-3	100 → 97	100 → 97	100 → 95	99 → 95	99 → 95
Surfaces	Agilent-3	43 → 44	35 → 46	47 → 46	46 → 55	46 → 53
Surfaces	Mira-1	53 → 72	59 → 73	50 → 63	56 → 66	55 → 68
Surfaces	Pendar-2	89 → 98	86 → 98	55 → 64	93 → 98	95 → 98
Surfaces	Pendar-3	73 → 72	67 → 70	69 → 64	72 → 70	72 → 70
Surfaces	RMX-2	55 → 90	62 → 87	73 → 88	67 → 81	68 → 83

Backup: remaining preprocessing experiments

Experiment	Doable?	Reason
Other normalizations (SNV, vector, area, derivative)	Yes	Classical models only, ≈42k fits; hours of computing
Wider spectral range (400–1,849 cm^{-1})	Yes	≈63k fits; hours
Robustness to added noise, background, spikes, shifts	Yes	Reuses trained models; mostly prediction
Reduced datasets (500 / 575 spectra)	Yes	Classical-only version is quick; with CNNs ≈539k fits
Preprocessing chosen from spectrum quality	Partly	Supported in 54 of 260 tests, all at CWA; full search ≈1.6M fits
Preprocessing chosen per instrument brand	No	Instruments were tied to stations: within a station, holding one out leaves $\leq$1 unit of that brand to learn from

Backup: how much data, details

Rule (Zhang & Gupta, 2000): N samples $\times$ M outputs give $N{\cdot}M$ constraints for W weights.
$N{\cdot}M > W$: overdetermined, one stable fit.
$N{\cdot}M < W$: underdetermined, many fits.

	W	$N{\cdot}M$	$W \div N{\cdot}M$
CNN, one station	208,691	54	$\approx$3,900
VAE, NATO data	$\approx$100k	27.5k	$\approx$3.6
VAE, Paper 2 design	20–80k	17.5–87k	$\approx$1

Modern methods (early stopping, convolutions, augmentation, pretraining, physics-based decoders) let compact models work below the classical limit. Our CNN was thousands of times underdetermined and still matched Random Forest — but did not beat it.

Confidence comes from the test. The 95% range of an 80% accuracy:

Independent test samples	Range
19 (today's substrate tests)	$\pm$18 points
60	$\pm$10 points
250	$\pm$5 points

Plan $\approx$300–500 crossed preparations, $\geq$60 independent test samples per key comparison and $\geq$5–6 instruments. Size the final study from a $\approx$100-preparation pilot learning curve.

Ratio above 1 = underdetermined. $M = 3$ for a three-chemical classifier; $\approx$46 for a spectrum (46 principal components hold 95% of the variation). VAE sizes are assumptions.

Backup: Paper 2 translation and ablations

E5 Instrument translation (needs $\geq$2 instruments): convert one instrument's spectrum into the other's; compare with the real paired spectrum and with piecewise direct standardization (PDS).

E6 What matters: remove the physics structure, the labels, or the analyte $\times$ substrate interaction term; learning curve versus number of preparations.

Backup: glossary

Balanced accuracy	Hit rate per chemical, averaged over chemicals. Chance is 33% for three chemicals.
Held-out instrument	An instrument whose spectra are used only for testing.
Independent sample	A separately prepared physical sample; repeat spectra of it are not independent.
Parameter	One adjustable number a model learns from data (a weight).
Extra Trees / Random Forest	Many decision trees voting; Extra Trees chooses its splits more randomly.
SVM	Support-vector machine: finds the boundary with the widest margin between chemicals.
CNN	Convolutional neural network: learns local spectral patterns such as peak shapes.
arPLS	Asymmetrically reweighted penalized least squares: estimates and subtracts a smooth background.
SG smoothing	Savitzky–Golay filter: local polynomial smoothing (11 points, degree 3).
Confusion matrix	Table of true versus predicted chemical.

Backup: references for Paper 1

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Backup: references for Paper 2

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